

BITS & PIECES

* I am thankful for the reader questionnaires that have been flowing in here from my mailing with the February issue. The response has been good. I will tabulate the results and try to print some generalizations, results, and trends next month.

Since this month marks the start of the third year of this newsletter, many of you have 24 for your last issue, which is this one. To avoid being removed from the mailing list, (and avoid a last minute overtime schedule here) please send in your renewal check (\$20 for 12 issue year) ASAP. This will be much appreciated as it will save me time and money in eventually tracking you down.

One item which surprised me on the questionnaire was the general consensus of H-SOOP readers that they would like to see little or no NEW STUFF! If you remember, in January I made a comment that NEW STUFF was taking over this newsletter, and I wanted to cut that column down. It appears most agree with me.

Since I feel it is important to keep abreast of new developments, and who has what, this column will never be discontinued, but I will begin to slowly phase it down to a more condensed form, with general headings of SOFTWARE, HARDWARE, MISCELLANEOUS items, and what have you. You'll get just enough information to know what is available and who has what. It will then be up to you to contact the vendors for additional info. I hope this doesn't upset vendors too much. I do not want this newsletter to turn into little more than a free advertising rag! Anyway, bear with me while I figure how to change to format, and slowly phase into it. I am formulating plans for direct mail advertising for vendors at a reasonable cost. I will be announcing more details of this next month.

* I sprung for an H-25 and am very happy with it. This, like most of the recent computer type Heathkits comes with the three main boards fully assembled and tested; there is only one board to assemble. It is a tiny board with 4 LED's and 2 IC's on it. The rest is all mechanical assembly. Following the instructions, it goes together quite nicely, with no tricky alignment procedures required. I only had problems with my print head going into an over-travel condition and triggering the fault light and causing the printer to shut down. It took me a long time to figure this out, as the "trouble-shooting" guide did not list that problem (or many others as far as that goes--very sparse, again). I finally figured it out and bent the over-travel arm out a bit so it wouldn't touch the microswitch till the very last instant, and that solved all my problems. A call to Heath Technical staff (when I could get through to them) confirmed this. It works like a charm, is a very well designed and operating printer, and works with HDOS and Magnolia CP/M with no modifications.

There are many switches on the back for setting things like page length, pitch, etc, the usual ones, and I feel they are in a bad place if you are constantly changing parameters for doing different work (which I am doing). You have to get around to the back and be fiddling with the tiny dip switches. I feel the more common functions could have been placed in the front under the cover or somewhere more accessible. I also would have liked to see a knob on the paper feed mechanism brought to the outside of the printer so one would be able to manually adjust and feed the paper. To do this you must either use the paper switch controls (on front panel) TOP OF FORM or FORMS ALIGN, or open the cover and turn the paper feed

pulleys. There are also dozens of flashing LED indicator lights on the internal circuit boards which are forever hidden from view. I like flashing lights, especially when they mean something. I'll probably eventually bring them up to a panel over the printer so they decorate the room.

The interesting thing, I went to the local Heath store to get a TI 810 because I wanted the 150CPS speed. After evaluating the two, and considering the self service and Heath parts, I figured this was the best way to go - even though they didn't have a printer stand, ribbons, cables, or paper in stock!

While on this subject, Larry Shipinski told me that the head drive belt, because of all the tension, will wear out prematurely if not lubricated. He suggested lubricating the teeth side with a proper grease type lubricant which will not hurt the rubber, such as LUBRIPLATE. I suppose Vaseline would also work.

* For those with an H14 printer, M. Graziosi tells me a much better ribbon, which is available at most office supply or stationary stores is the B-72 from Burroughs. It works on the H14, and is black in color. He also tells me that the MM5257 (TMS4044) which is a 4K X 1 RAM is available for a great price from Digital Research: Parts/ POB 401247B/ Garland, TX 75040/ (214) 271-2461. The listed price is \$2.95 for the 450 ns version. Mario was told this should work with the Heath 8K and 16K boards.

* Tom Lee writes in to give me additional info on the Intra connector mentioned in the December and January issues of this newsletter. I mentioned them for bringing out two disk controller cables from one disk controller card. Tom mentioned that since I did not supply a part number he had a hard time getting the right part. Tom says the correct part number is an AP Products part, Model IC-34, AP 922567-34-R INTRA-CONNECTOR. I hope this info makes it easier to obtain the proper part.

* For FLOPPY DISK REPAIRS, an outfit called COMPUTER SERVICE CENTER/ 1023 N. LaBrea/ Hollywood, CA 90038/ (213) 851-2226 advertised in Computer Shopper a three day average turn around time for repair of disk drives, including Shugart, Qume, Tandon, Siemens, MPI and others. Right below this add was another by The Micro Computer Repair Center/ 9 Main St/ Danbury, CT 06810/ (203) 797-9997. They have a 72 hour turn around time on most items, with 30 day guarantee. Some sample repair prices for time, parts extra, are \$29.95 for 5" disk drives, \$49.95 for 8" drives, and \$99.95 for hard disks. Yet another, appearing in the same Computer Shopper is PDS/ 2670 Walnut Ave, Suite C/ Tustin, CA 92680. They sell and repair all major brand drives. This info, with what I published last year, should give you plenty of drive repair places to check out. I have no first hand info on any of them, so don't call me for advice. Good Luck!

* Another place I gleaned out of Computer Shopper is called Dealin' (415) 493-5930. Their add says they have friction and pin feed platens, HY-II extender cards, power supplies, cases, tractors, keyboards, feeders, stepper motors, etc., for DIABLO PRINTERS. Contact them for more info, but be aware that I already have a two complaints against them for not supplying as they advertise.

* Some of you have written on your questionnaire for more information on PASCAL NEWS. It is a non profit organization, and because of this, their service is very

slow and unpredictable. To order, contact Pascal Users Group/ POB 4406/ Allentown, PA 18105. Rates are \$10 for one year, \$18 for two, and \$25 for three. It is scheduled to come out 4 times a year. The address I have listed I took from a recent mailing I received and I don't know how accurate it is. When I said they are slow, I meant it. I sent them a bunch of money about a year ago, and thus far, have only received my 1979 issues. The address I used when I ordered was: Pascal User's Group/ % Rick Shaw/ POB 888524/ Atlanta, GA 30338. If anyone has more current info, and finds these listed addresses incorrect, please contact me so I can correct this in the next issue of this newsletter and write to them about my missing last 2 years issues!

* Some clarification is needed on the CP/M BIOS-80 software. It is important to note that there are now two versions. The first release supports ONLY Heath CP/M 2.2.02, and ONLY the single side, single track drives, AND the double track double sided drives - nothing in between. It is still available and sells for \$30. With this version you must "make up" the BIOS using auto submit files supplied. It's not hard, but takes awhile. This was done to avoid legal problems with releasing a full BIOS. There are no planned updates to this version. For those interested using it with the Heath Z80 CPU, see Technical Forum. In the future, a patch may be published to allow this version to work at 4Mhz.

The latest version available is for CP/M 2.2.03 from Heath and sells for \$35. It supports ALL drives from the 100K to 400K and anything in between. It also comes complete with BIOS, no more of this submit hassle. This means you are up and running a few minutes after opening the package and configuring your system. It will not work with 2.2.02. This version will support the Heath Z80 card, while the 2.2.02 version will not. Those who have purchased the 2.2.02 version can get an update for \$10 to the new version. If you are ordering from myself, the author Ray Livingston, or anybody else, please specify the version you want. For updates, be sure to include the original distribution disk. A 4 Mhz update is now being worked on and will be available in the near future.

* Ray informed me that Livingston Logic Labs will not be releasing the H8 double density 8" controller card because he did not receive enough feedback to indicate this would be a profitable venture. I told you in previous issues to write Ray if you were interested - well, hardly no one did. On the positive side, CDR will be releasing the 8" DD controller for the H8 and Ray will be doing the HDOS driver for it.

Ray also told me that his 5" DD version of the controller cards for the H8 and H89 will be available late March. The schedule has slipped a bit, but it is coming out.

* For those wanting to use Ray's 8" controllers on the H8 you must have a Z80 CPU installed. For those who would like to have 8" SD but not wanting to purchase a Z80 from Heath, D-G, or Trionyx, there is another alternative. That is to modify your present 8080 card using the mod by Steve Tinius as reported in issue 14 page 7 of this newsletter. Steve is making a special deal to those purchasing the controller from Ray. He will deduct 10% off the published prices as follows: Bare board and parts list \$30; Complete kit \$75; Assembled and tested board for \$90. This board simply plugs into some IC sockets in the 8080 and is easy to install. For more info on this board, or to order, contact Steve Tinius/ HQ USEUCOM Box 741/ APO NY 09128.

* While on this subject, Heath new 2.2.03 CP/M will not boot with the old monitor ROM set. You will need the 0-org CP/M ROM set. If you have just installed the address decoder ROM, it won't work. However, there is a way to make it work without the new ROM set. Boot 2.0 HDOS and reset your system, but don't power down. This sets a bit in memory. Then if you boot the new CP/M 2.2.03, it

should come up.

* Jim Benesch reports an error in the Technical Forum, H8 CRASH FIXES as published in the February issue (pages 7-8). The article says "Then I replaced the masked ROM from Heath, ordered to replace the slower EPROM sent with the Extended Configuration Option." The wording as I published it was not clear. It should say "Then I received the masked ROM from Heath (P/N 444-70, about \$16 from parts department) and replaced the EPROM supplied with the Extended Configuration Option with the masked ROM."

* Some interesting information was called to my attention by several readers in regard to **auto:boot on power up** for the H8 and H89. I did some investigating and here's the scoop! If you have the MTR-89 monitor chip installed (which H89 FA's, upgraded H89's and H8 extended configuration option have) the procedure is simple. Although Heath does not really document this feature, it is there. It was documented for the H8 extended configuration option. If you look through the source code for the MTR-89 monitor ROM, you'll also see it there. Switch 501 on the H89 CPU and H8 extended configuration board has switch 7 which most documentation says either "undefined", or "for 9600 Baud operation". Switch position 6 is for the baud rate, 7 is for auto boot on power up. If this switch is set for "1" (Normal position is "0") the system will boot on power up. (See page 14 of MTR-89 manual, page 5-3 of Z89 operation manual, or H8 extended configuration manual). Setting this switch position to "1" will cause auto boot on power up or on reset. Try it and see. If you are using the Magnolia ROM, it is again position 7. With the Ultimeth ROM's, it's position 6.

* H37 users, the update software necessary to run your H37 DD card should now be available from your Heath center (it also contains an H25 and MX80 driver). There always was a driver on your test disk, but it was disguised. To run the new H37 software, you will need a new monitor ROM, MONITOR-90. If you don't have it installed the system will hang when you do an INIT. This info from Larry Shipinski.

* This item is well worth the subscription of REMark alone, if you are running MBASIC version 4.82. As those of you know who are using it, it takes more memory than the previous version. Many program that would run on the 4.7 release will no longer run on 4.82 since more memory is used. The February issue of REMark, page 18 provided information on this. The 4.82 version loads both overlays of HDOS into memory, thus it requires more memory. The HDOSOVLO.SYS is required for commands such as OPEN, CLOSE, FILES, etc, and HDOSOVLI.sys is required for RESET. If you do not use the RESET command (which I never do anyway) you can patch MBASIC 4.82 so it only loads the first overlay into memory. To do this, use PATCH as follows:

Change all the following address locations from 160241 through 160247 to 0. The old data values are 076, 001, 377, 010, 322, 217, and 152 respectively. You must use the PATCH program supplied with HDOS 2.0. This should solve lot of your memory problems.

* A new release of SPELLBINDER is available. Those interested in updates, \$50 plus shipping is the update charge, which includes new manual and disk. The old spellbinder was great, this one has more features and better supports proportional spacing TABS and other functions. I won't get a change to play with it till after this newsletter, if anybody wants more information, give me a call.

* JIM BLAKE-- Welcome back to HUG! Speaking of JB and the NATIONAL HUG HEATH ZENITH meeting, JB called to give me tentative info just at press time. The dates set aside are August 6-8. It will be held at the Regency Hotel at O-Hare in Chicago. Rates are \$15 registration (get this to JB ASAP). Hotel rooms are \$16.70

for a double. Let JB know who is interested so they can plan accordingly. Vendors will receive free exhibit space.

Briefly, the tentative agenda is as follows: Friday noon and evening is registration. Saturday morning is opening remark with Bill Johnson, President of Heath. 10:30 to 12:30 is clubs presentations. After a lunch break, 1:30 to 3:00 is sneak previews. Barry Watzman will take the floor next, till 6:30 for Question and answers. 6:30 will be cocktail party and dinner, with Don Moffet of ZDS speaker. Sunday morning is free time, Noon is classified, and it plans to be over at 4PM. There is a chance that those remaining Monday will be flown to the St. Joseph, Michigan Heath plant for a tour, and returned to Chicago.

It sounds like they just about own the hotel for that weekend. Sounds exciting! I'm planning to attend and meet a lot of you.

NEW STUFF

As mentioned in BITS & PIECES, the format of this section is being changed. Vendors, in the future please supply me with a short paragraph to say what you want to say. Include all important points and essentials in the paragraph to give the reader a good idea of what you have. If you want to send me additional information, fine, but for this column prepare a short column. If you don't, and I have to do it myself, it may not say what you would have wanted, and that will be your fault. I will try to print what you send me as close as possible, but reserve the right to edit all materials.

SOFTWARE RELATED

* Hoyle & Hoyle Software/ 716 S. Elam Av/ Greensboro, NC 27403 has new prices on their products. **Steer Kleer** which is four obstacle courses sells for \$14. **A Remarkable Experience** is now \$16, and **A Galactic Experience** is \$25. Now available is complete map sets and hints for \$10 per game. To receive their burp sheet on their software products, contact them.-- A word of congratulations again goes out to Don Harton who was the first explorer to complete Hoyle and Hoyle's second game, **A Galactic Experience**.

* The ACGNJ (Amateur Computer Group of New Jersey) is starting a Special Interest Group on **CP/M (SIG/M)**. The SIG/M group is serving as secretariat for the distribution of public domain software in the same manner as the CP/M users group used to do before LIFEBOAT associates made a commercial enterprise out of it. It is the intent that the software released by SIG/M be freely available and copyable by anyone for use in NON COMMERCIAL purposes. In other words, you or your club (or group) could get a copy (nominal copying charge) and make as many as you want for friends and not be violating any copyright laws! (Jump for joy, all you PIRATES).

There are presently 52 volumes available, which cover just about everything from simple games, to remote bulletin board systems--free! Although the common distribution format will be 8" SS SD, librarians have been appointed to support other formats. Robert Todd Jr/ 1121 Briarwood/ Bensalem PA 19020/ (215) 752-4604 is the Heath librarian and is converting and making available all CPMUG software on Heath compatible diskettes. For standard 8" disks of the SIG/M variety, Robert suggests the SIG/M be contacted directly at SIG-M/ Box 97/ Iselin, NJ 08830. Robert will handle the CPMUG disks.

Robert will support Heath standard 40 track hard sector SS, 80TK DS Tandon format, Heath soft sector DD 80TK DS (H-37 controller compatible), and Heath soft sector 40TK SS. Tentative pricing is \$6 per volume which includes shipping and handling, UPS addresses only. Add \$1 for US mail addresses, and foreign orders add \$3 extra.

This is for formats which will only need one disk, such as the 400K Tandons, and H37 DD DS. For 40 Track SS 10 hole hard sector, since it will average 2 to 3 disks to hold the 8" volume, the price is \$16. Note these prices include disk media. Also note that files larger than 90K will not be able to be copied to the 40 Track 10 hole hard sector. For more information on this, or to order, contact Robert Todd.

* **STAR MERCHANT** is a game of interstellar trading that runs on a 32K H8/H19 cassette operating system, under Extended Benton Harbor Basic (10.6.0). Can you become a billionaire in 10 years? Cassette copies are available for \$10 (plus California sales tax where applicable). Order from Kurt Schultz of THIN KING THINGIES/ 155-1 Roxanne Ct/ Walnut Creek, CA 94596. Kurt says it may run on an H88, but since he doesn't have one, he does not know.

* **Generic Software/ POB 1154/ Troy, MI 48099/ (313) 879-1115** has an enhanced WH-24 line printer driver called **E24DVD**. It maintains all features supported by the standard Heath driver for the WH-24 (TI-810), plus a greatly expanded HELP option. The HELP option displays the range of accepted values, the default value, and current value for each device parameter. Price is \$15.95 which includes 5" distribution disk which contains the executable .DVD file and an .ASM source file. The price also includes shipping and handling. Michigan residents add 4% sales tax.

* They also have a **DATABASE EDITOR** called **SEQ-BASE** for \$29.95. It is a menu driven, comprehensive system of programs to create and maintain user-defined database files. Graphics on the H19 can be used to help the user define the desired formats of input and output. Search can be done on any field. There is a fast assembly language Shell-Metzner sort routine, selectable to sort on any field.

Output can be provided for generation of labels and reports, outputting the entire database or a portion as selected. It uses sequential disk files, but is a memory based system, therefore you are limited to the number of records you can handle at a time. Figure 100-75 byte records for 48K HDOS, or 300 for 64K CP/M. **SEQ-BASE** requires H89 or H8/H19 with minimum 48K RAM for HDOS, or 56K for CP/M. An MBASIC interpreter is also required in addition to the operating system. Specify CP/M or HDOS version.

* **CXREF** is a utility which will create a sorted cross reference of a C/80 program. It will optionally produce a line numbered listing of the C/80 program followed by a cross reference of all the symbols used in the program. All variables and function names will be displayed up to a max of 16 characters in length. Available in ABS format on 5 1/4" disks for HDOS 1.6 and 2.0. Price is \$14.95. Order or obtain more info from Brian Lockery of **Interactive Micro Systems/ POB 21007/ Columbus, OH 43221/ (614) 363-1534**.

* **Bill Hall of MLM Associates/ 5621 Maple Heights Ct/ Pittsburgh, PA 15232** has some easy interface modifications to HUG's MORSE8 program for use with the 8250 ACE on the H89 serial interface board, or H8-4 card. If you have a copy of MORSE8 on disk or cassette and want to reduce your interface problems send MLM a blank disk and return postage and they will provide you with a modified version of the code. \$4.50 if you do not want to send disk. \$3.00 for printout listing for cassette users. For more info on this interfacing scheme, contact MLM.

They also have available **ARCHIVE**, a dynamic handler allowing you to manipulate groups of files. It runs under any current version of HDOS, and allows you to create an archive file, update existing files, add files, extract files to any device, delete files, and list file names by title in

the archive. One of its most important uses is to store many small files on the high density drives in an efficient manner. It was written in RATFOR and comes with source code. Price is \$19.95, distributed on 5.25" 10 hole hard sector.

* **PEDIGREE.BAS** is an MBASIC pedigree management system for animal breeding purposes. It comes in a version for dogs, cats, and general utility. The purpose of this program is to print out a three generation pedigree which is useful for record keeping; for sending to individuals interested in buying your animals; or for visualizing the pedigree resulting from various breeding combinations. The program is menu driven with an on-screen form to promote easy data entry. System requirements are H89 or H8/H19 with 1 drive minimum and 48K, along with a printer capable of printing 132 character line. The program is supplied on 10 hole hard sector disk for \$20, specify HDOS (will run on 1.6 or 2.0) or CP/M (2.2). Order or obtain more info from **ROBERT KOESTLER**/ 640 Trephanny Ln/ Wayne, PA 19087/ (215) 687-3194.

* **APOGEE SOFTWARE**/ Box 15124/ Savannah, GA 31406 has several software packages available. **UNERASE** is an erased file recovery utility for CP/M 2.2.02, while **UNDELETE** is a deleted file recovery utility for HDOS 1.5 and up. These are both for the 100K drives and sell for \$14.95 each. **MAZERACE** is a graphics game for 32K HDOS and H89 or H8 with H19 and H17. This is a timed game where you must race through a maze (you can create your own) and reach all the goals in order. It sells for \$12.95. **STATPTCH** is a STAT command mod for CP/M 2.2.02 which automatically modifies STAT.COM to provide you with information on your disk drive performance. Also included is **ERRORS.COM** which displays the error count. For 2.2.02 only, it sells for \$12.95.

Add \$1.50 postage on all orders, Georgia residents add 4%. All software supplied on standard 5.25" hard sector SD disks.

* From Lawrence Fina comes **JOB COMMAND UTILITY** package. This improved Job Command Utility is a 2K program that takes SUBMIT or DOCOM type programs one step further. **CDS.ABS** allows you to store several Submit type arguments in an ASCII file. **CDS.ABS** then takes a user's input code and matches and decodes it, then feeds it into the type ahead buffer and exits. HDOS takes over and reads the buffer as if you typed in the output argument yourself. Cost is \$10 for the .ABS and documentation, or \$30 which includes source code.

Lawrence also has **UTILITIES DISK** for \$8 (.ABS) or \$15 with source. It includes the following: An improved **NEW DATE PROGRAM**. A **TRANSMIT** program to transmit H89 screen to the LP after each CR. **COUNTS** an ASCII file by bytes, lines, sectors and 57 page passes, and another version for 85 lines per page. **SAFE** disk reset program with messages and error processing (similar to a program seen in REMark). **DISK** changes disk motor turn off time for Heath's old SY.DVD to 5 seconds (also seen in REMark).

I have used the job command program and find it useful for larger types of handling than a SUBMIT will handle, or for combine several SUBMIT type files. For more info or to order contact Larry Fina/ 16144 Sunset Blvd Pacific Palisades, CA 90272/ (213) 454-6393 evenings. For \$2 you can get program descriptions only. Add \$6 for disk, or send one. Add \$2 per order postage. First update at no charge.

HARDWARE RELATED

* Update from **RAY ALBREKTSON**/ 619 W Dover Dr/ San Bernardino, CA 92407. Tested and ready to go **CLOCK/ CALENDAR/ SOUND** boards for the H89 are available for \$85. Bare boards with instructions for \$22. A full disk of software for the board, including auto prologue that

automatically updates system dates on boot-up for \$5. It is compatible with Steve Robbins 'TIME FOR HDOS 2.0' package. The **AUTO BOOTER** kits mentioned in February H-SCOOP are complete kits for \$22.

* The **PC-89** will be available early March for \$224.95 (introductory price) for an assembled and tested board including support disk and doc. This was written up in past issues of this newsletter. Briefly, it is a perpetual clock/ calendar board with 9 port parallel interface for the H89, using the versatile and popular 8255A PPI chip. The board includes self charging Nicad battery for clock and can be installed in the LEFT side of the H89. To order, or for more info, contact Don Long or **ENVIRONMENTAL CONTROL SYSTEMS**/ 9319 Willowview Ln/ Houston, TX 77080/ (713) 464-1717.

* Update from **LIVINGSTON LOGIC LABS**/ POB 5334/ Pasadena, CA 91107. The prices of Ray's FDC series 8" SD controllers have been reduced and are only being sold as assembled and tested units for \$200 for the H89 version, and \$225 for the H8 version. These include software for CP/M 2.2.02 or .03 at no extra charge. The HDOS driver is available for an additional \$35. All Rays products are now being shipped with CP/M software only, to avoid confusion.

The new release of **BIOS-80** for CP/M 2.2.03 is available for \$35, or \$10 update charge with return of version 2.2.02. Available from myself, Ray, and Keyboard Studio. Refer to BITS & PEICES for description. There will be a 4Mhz version also available. Same price, \$10 for the 8" updates to 2.2.03, get these from Ray.

The **PC-17 DD 5"** hard sector controller boards should be available in early March. Orders are now being accepted by Ray. Price is \$275 for H89 version, and \$300 for H8 version. Ray tells me he will make group offers available to those buying in quantities of 5 or more. Ray will also be releasing a clock calendar board for the H89 in April.

I have also received my H89 extender card from Ray and it's great. Cost is \$35. It allows you to get the small bus boards out to work on them.

* **MICROFLASH Co.**/ 4916 B Carol/ Skokie, IL 60077/ (312) 677-4928 will soon have an **H89 EXPANSION BOX** available, the M-89. I have had many individuals tell me that, but this is the first time I have actually seen it in print. The box will have 9 board locations and be fully compatible with the H89 I/O Bus. They are also working on PC boards for this interface box which will include an X-Y plotter, AC controller (for studio), and voice synthesis to name a few.

Microflash also has available an H-89-1, which is a decoder interface for \$49.95 assembled, H-89-2 parallel I/O card for \$79.95 assembled, H-89-3 EPROM burner for \$79.95 assembled, and H-89-4 Extender board for \$34.95. Refer back to issue #15 for a description on these first three items. Write MICROFLASH for more information.

* Just at press time: **Magnolia Microsystems**/ 2264-15th Ave West/ Seattle, WA 98119 will be releasing mid March an **INVISIBLE DISK** which tripples RAM for H89's. It allows up to 176K RAM to be addressed by the system (H88, H89, H90's) used as a disk drive. You load CP/M or HDOS, load your programs and run them at incredible speed. The RAM board frees your built-in disk drive to handle files and data. It can also support MP/M for up to three multi users. The CP/M is handled via Magnolia CP/M BIOS, and HDOS is handled by a driver written by Dean Gibson of UltiMeth corp. This board replaces the 16K expansion board on the left side of the H89. Price for the 128K RAM board (uses 4164's) is \$595.

Also from Magnolia (answer to a REQUEST) is **VIDEO OUT** from H89's (and H19's terminal and computer types).

The board permits the CRT display to be reproduced on a television monitor or recorded on videotape. The video out comes through a standard BNC type connector and can be used with any high resolution monitor or VTR. Delivery in mid March, price \$79. Both products will be on display at the West Coast Computer Fair in San Francisco, March 19-21.

MISCELLANEOUS

From Compulink/ 1840 Industrial Circle, Dept A/ Longmont, CO 80501 (303) 651-2014 comes SÓOPER SPOOLER, an intelligent printer interface for \$349 with 16K memory and parallel interface. Add \$95 for serial board and \$159 for an additional 46K memory. What this device is, is an electronic spooler. You connect it between computer and printer and output to printer like you normally do. This device has buffers to hold the information and transfer it to the printer as the printer can handle it. The computer fills this up (thinking it's a printer) and can return to it's computing while this thing is outputting to printer.

For \$845 you can have a Smith-Corona TP-1 daisy wheel text printer. The add I saw tries to make it sound good by saying it prints at 120 words/ minute. If a word is 5 characters, that would be 600 characters per minute, or 10 per second, which is real slow! But for that price, you may want to check it out. It's available from Micro-Printer Marketing/ (215) 433-3366. No address given!

Available from Greg Marsh/ 5279 Miles Ct/ Woodbridge, VA 22193/ (703) 590-3360 (7pm till midnight) is a six drive update for HDOS. Greg's instructions, along with the new DKH17V2.DVD from UltiMeth (or myself for \$40) will allow you to run up to six 5 1/4" drives with an H8 or H89. He also has the CPM BIOS mod to handle 6 drives as well. Call him for availability of the CP/M version. The HDOS version is ready and runs under HDOS 2.0. Info is supplied on 40TK SS disk for \$20.00. I looked through the documentation and it appears to be clear and easy to follow. A minor change to be able to run 4 or 6 drives. Four sounds like a good number since that would allow you to run 2 100K and 2 400K drives, for maximum compatibility.

I also understand Greg has available quite a nice comprehensive STAT program for use with HDOS. If you are in need of this type of program, ask Greg to fill you in.

BIBLIOGRAPHY UPDATE

February: 1982 Microcomputing, page 78 carried a one page article "GET CONTROL OF YOUR CURSOR by Joseph Hesse. It deals with using Microsoft FORTRAN to control the H89 terminal functions.

REQUESTS

J. D. Smith/ 5100 E. Tropicana, Apt 478/ Las Vegas, NV 89122 wants to know if anybody in H89 land has used their H89 with a Dental Software package, or knows of any dental software programs that will run on the H89. James is a dentist with an H89 and wants to put it to work in his profession. If you have or know of such a package, contact James.

This is a problem that came up long ago, and there was an answer, but I forgot what it was. I should not have forgot, since I have a Diablo 630, but here is the request for anybody that can help. Jim Benesch/ POB 1811/ Minden, NV 89423 writes "Has anyone successfully found a cure for an apparent bug in the Diablo 630 firmware which deletes a tab on the top of page after the printer produces a form-feed? This is very disturbing when printing source code. I have contacted Diablo, tried new printers, and even installed the word processing ROMs without success." If you have the answer, write or call Jim. His

phone is (702) 266-3686 evenings or weekends.

Alfred Llorens/ 605 East Rockcreek Dr/ Columbia, MO 65201 needs information on how to connect Heath's X-Y plotter to an H89. Al especially needs to have a Digital to Analog converter circuit along with instructions on interfacing it, preferably to an RS-232 serial port. Any info, contact Al, or phone at (314) 445-7558 after 5 PM CST.

Chris Simmons/ 2802 N. Greenwood St/ La Grande, OR 97850 is interested in knowing if anyone knows how to interface the H19 to a remote display. He points out that REMark issue 6 page 16 had an article on interfacing the H9 to a remote. He tried to locate the points on the H19 and tried that without much success. He got 3 cursors on the screen, one at top, one at bottom, and one in the middle. These cursors bounced up and down. Anybody have the scoop on this, contact Chris or myself, as I feel it would be good to publish this for the benefit of all.

WHO'S WHO

Ted Benglen is interested in forming a Northern Colorado HUG (Ft. HUG). They will be meeting at least once a month in the Fort Collins area and serve users in the Loveland, Greeley, Longmont, and Boulder areas. If you are interested, or wish to request additional information, contact Ted Benglen II/ 822 E. Country Rd. 30/ Ft. Collins, CO 80525. Phone (303) 669-4116. Ted, by the way, is the president of MI-8, the ones with the talking H8 among other things.

There is a newly formed PASCAL MT+ USER GROUP (MTPUG) promoting the use of Pascal as a programming language, and serving as a vehicle for communications between users and the language. Membership dues of \$7 US, \$8 Canada and Mexico, and \$10 foreign surface mail (\$16 air mail) will get you a quarterly newsletter with bug reports and fixes, programs, questions and answers, and items of interest. For more information contact MTPUG/ POB 192/ Westmont, IL 60559. In Europe contact MTPUG Europe/ Schimmelmännstr/ 37A/ D-2070 Ahrensburg/ WEST GERMANY.

* PROGRAMMING GOODIES *

In last months issue I printed a program to DRAW A BOX (FRAME IT) taken from HUGNJ newsletter. There are some bugs that should be noted caused by the author's switching from BH BASIC to MBASIC which were not upgraded.

To allow this program to properly run under MBASIC, simply place PRINT statements before each "cursor position" statement in lines 200 and 310. While you're installing PRINT statements, add two at the end of the program to prevent BASIC from erasing the bottom of the box with its ending comment. These corrections taken from December 1981 HUGNJ. I apologize for any problems this may have caused. Because of my lack of time, this is one program I did not try before printing it.

H8LEDS QUICKIE

Oren Wood sent in this quickie to turn on and off the H8 Front Panel LEDS.

```
10 REM   TURN ON H8 LEDS
20 POKE 8200, 131
30 REM   MESSAGE FOR LEDS
40 FOR I=0 TO 8
50 READ A
60 POKE 8203+I,A
70 NEXT I
80 DATA 146,140,143,143,129,255,144,143,143
```

```

90 LINE INPUT "DO YOU WANT TO TURN OFF
LED'S ? ";A$
100 IF A$="YES" THEN POKE 8200,195
110 END

```

H8 PAD POLLER

As long as we're on the H8, I may as well throw in a few more. This one from Larry Carlson is an Assembly Listing of polling the H8 keypad. Some of it is from the PAM8 listing, and some is Larry's. The problem with simply calling RCK from PAM8 (Read Console Keypad 003260A) is that it does not return until a key is pushed. In other words, it will sit in this routine and hang up until a key is pushed. It is many times desirable to be able to poll the pad, but not stop the program until a key is pushed.

```

*
* H8PAD POLES.THE H8 PAD TO SEE IF KEY
* PUSHED RETURNS WITH BINARY IN "A",
* ZERO RESET IF KEY PUSHED, RETURNS ZERO
* SET IF NO KEY IS PUSHED, RETURNS CARRY
* SET IF NUMBER IS NOT OCTAL (GREATER
* THAN 7). NO DEBOUNCE. USES A, F.
*

```

H8.PAD

```

IN      360Q   *INPUT H8 PAD
CPI      377Q   *NO BUTTON DOWN
RZ
ANA      A      *CLEAR FLAGS
XRI      376Q   *INVERT ALL BUT
              "0" BIT
RRC
RNC
XRI      010Q   *0-7 HIT
RAR
RAR
RAR
RAR
RAR      *ROTATE 4X RIGHT
RET

```

(+=10, -=11, *=12, /=13, #=14, .=15)

Larry brings up an interesting point. "The Z80 processor is great I'm sure, BUT there are many reasons to continue assembly in 8080A. They are Intel's 2 new CPU's, the 8088 (IBM Home Computer) and 8086 (Heathkits next 16 bit Computer?). Although these have 16 bit instruction sets they are compatible at the assembly source code with 8080A. Want to write a program for any computer using 8080A, 8085, 8088, 8086, or Z80 CPU? Use 8080A assembly."

CP/M DDT RELOCATOR

This article was taken verbatim from December 1981 issue of HUGNJ, written by L. Shepard.

"Credit for the following helpful hint is given to ACGJN and other nameless souls who have delved into the innards of this operating system.

"DDT.COM, as most of you know, is a powerful and flexible program which allows dynamic interactive testing and debugging of programs generated in the CP/M environment. The program is initiated in one of several ways, but typically by typing its name and the name of a file to be debugged, i.e. "DDT FILENAME.COM/ or .HEX/".

"Briefly, operation begins when the first sector of DDT is read into memory at 0100H. This first sector contains relocating instructions which cause the balance of the program to locate in high memory, and in fact overwrites the CCP (Console Command Processor). Normally this does not cause any problems. (I suspect that Digital Research intended this overwriting to help protect the proprietary nature of the CCP). In any case once relocated, the TPA

(Transient Program Area) is left open to receive the program named in the call statement and operation of the debugging tool begins.

"Suppose however, that you want to patch some changes into your operating system and you need to use DDT to disassemble and examine various parts of the new system - in particular addresses within the CCP. From my comments above you will note that the standard DDT and CCP are mutually exclusive and therefore DDT cannot be used to look at the lower end of the operating system. Suppose further that DDT could be instructed to relocate at a lower memory location and thereby not overwrite the CCP. We are now at the point for this little article.

"DDT.COM can be modified to relocate in lower memory (8 pages lower) by a process so simple as to be guilty of a crime for failure to pass it along. If you could look at the first sector of DDT you would find coding that looks something like this. .14BH, LX17, MOV A,M DCA. These are the instructions that set the relocation address, changing them on the disk is a little difficult, but by following the directives below you can create a modified DDT that will locate 8 pages lower in memory.

DDT DDT.COM

NEXT PC (response after DDT sign-on)
NNNN 0100

```

-A14B LDA 7 (to assemble at named add.)
-14E SUI 9
C/R (denotes CR or empty line)
GO
At the A prompt,
SAVE 19 SPLDDT.COM (or whatever)

```

That's all there is to it, now you have a new DDT that can be used to examine your CCP and memory above." Many thanks to HUGNJ. This sounds like a good programming goodie for you CP/M hackers!

HDOS MBASIC PROBLEM?

Larry Wier writes in "It looks like HDOS MBASIC has a bug which has caused me a lot of trouble in the past. For example, if you type the following:

```

10 FOR I=1 TO 10
20 PRINT "HELLO ____"
30 NEXT I

```

"In line 20, when it is listed, only 'HEL' will show up. Nice HUH! Anyway the problem seems to be in the EDIT function since the EDIT function interprets the underscore character as a DELETE. It seems strange that the preceding quote (") doesn't seem to work in this case.

"The problem does not exist with CP/M in this form. It may still be there in some other form but I haven't found it yet." Any feedback on this, Larry can be contacted at 1068 149 Pl S.E./ Bellevue, WA 98007.

TECHNICAL FORUM

MORE ON MX-80 with CP/M

Clem Pepper/ 3270-96 Caminito East Bluff/ La Jolla, CA 92037 wrote me up some info on getting the MX-80 to work under CP/M. I have published several hints on the MX-80, see Issues 13-7, 16-6, 17-4, 18-4 and 5, 19-2, 20-6 & 7. There usually has to be a change in cable or appropriate software driver if using on both HDOS and CP/M--ed.

"When one purchases an MX-80 for use with the H89 one must also obtain a serial interface unless a parallel interface has been obtained elsewhere. Epson provides a

special circuit board for the serial interface; their number 4181. It contains an eight position DIP switch and some wired in jumpers. If the switch position 1 is ON and the remaining 7 OFF the interface will be configured for 9600 baud, 8-bit word length, and odd parity check--which is meaningless because the next switch OFF disables the parity check.

"Before doing anything with this board the user should read the manual carefully, and in particular familiarize himself with the two DIP switches on the board above which the serial board is to be replaced. In particular, make sure the automatic injection of line feeds is OFF unless you want all your text double spaced. It is a real nuisance to get back to this board afterward.

"Returning to the serial board there are two jumpers to be concerned with. The first is labeled JNOR. It must be cut. Adjacent to it on the board are the two connection points for a jumper labelled JREV. A jumper must be connected here. After ensuring the board is securely installed and the ground wire connected re-assemble the unit, there is no more to be done here.

"This exercise with the jumpers is what I was given by the Heath store. I was also told that Heath does have a new BIOS, and if I should want to use both HDOS and CP/M I should go that route. For as far as I can see, CP/M is what I will be using, so I elected the hardware route.

"Now for the second half. One must now remove the top cover from the H89 and locate connector P603 on the serial board. It is at the very top. Solder a jumper wire between pins 7 and 14 of this connector. The best location it seemed to me is right at the bend of the pins going into the board. Don't overheat, but make sure to make a good, solid connection. (I choose to do this in the MX-80 itself--ed). A pursual of Heath's circuitry will show that this connects the DSR and CTS inputs to the INS8250-B ACE. Epson does not use the CTS function. These connections are pins 4 and 20 in the cable. To avoid messing up either the Epson or the Cable it seemed preferable to make connection on the serial board. (The problem with this is if you use another printer at times, it may not work, that's why I choose to do it on the Epson--ed).

"One more point of utmost importance. The system configuration must conform with respect to baud rate and the identity of the LIST device. So call up CONFIGUR, key "A", then "C" and type in 9600 for baud and 340Q for location. Then key "Y" for exit. Then key in C for default I/O configuration, which should be LST=LPT, at least that is the preferred setting for my unit. I don't know if it has been my doing or the machines, but a couple of times now in just a few days the baud rate has changed on me-- quite possibly from booting up from a different disk without thinking. The results were a lot of wild hair pulling while trying to figure out what could have gone wrong. So now, if and when the thing refuses to print, this shall be my first point-of-call.

H89 64K MEMORY EXPANSION--CHEAP

This info outlines a "cheap" way to add the 64K RAM expansion (H-88-16) to an H89. Since it is not available in kit form, you will need to order the following from Heath (or whoever).

85-2539-1 PC BOARD \$20
432-1074 10 pin edge connector \$2.15
432-1076 25 pin edge connector \$4.30
204-2518 Accessory mounting bracket \$7
266-944 Nylon guide (2 needed) \$.50

The total of these parts is \$33.95 plus shipping. (If you have your computer stationary you probably don't need the last two items for an \$8 savings. I have the Magnolia 64K add on in one of my systems and there is no bracket.--ed) You may also want to order the 16 pin dup adaptor

cable (134-1130), although the author used a short cable with a 16 pin dip connector on one end and soldered the other wires directly to the PC board.

The remaining parts can be purchased from JDR Microdevices/ 1224 S Bascom Av/ San Jose, CA 95128/ (408) 995-5430 (i.e. sockets, 4116 memory chips, 74S132, 79Ls, .01uf bypass caps, and the 10uf electrolytics) or any other reputable supplier for about \$25. You will need to borrow a copy of Heath's documentation for the WH-88-6 board. The only special note is to replace U562 on the CPU with a 74S132 chip. The chip (4116) at U549 is removed from the CPU card and placed on the 64K expansion card, and the interconnect cable plugs into the U549 socket.

Depending on the version CPU card you may need to use a jumper between JJ503 and the expansion board. This jumper is supplied with the H88-7 PROM upgrade kit. This procedure sure beats the Heath price of \$150 and works as well. For those interested Magnolia also supplies this board, which is smaller than the Heath board and does not require bracket. If anyone is interested I can supply them with that for \$125, shipping and handling included.

HA-8-6 Z80 ON BIOS80

Although Ray Livingston is not going to upgrading the BIOS-80 2.2.02 version anymore, several individuals have questioned me about running it with the Heath Z80 card. Ray called my attention at a patch provided in CHUG newsletter, December 1981 issue page 17. This patch was designed and tested by John Stetson, Jim Meek, and Dan Seely. It is for use with the H8 HA-8-6 CPU card.

"The following procedure details the steps required to patch Ray Livingston's BIOS80 for Heath CP/M V2.2(.02) to enable it to boot from 400K Tandon Disk Drives when used with the Heath HA-8-6 Z80 CPU card for the Heath H8 computer:"

From Prompt, DDT MOVCPM80.COM
3400 0100

-L2CD2
2CD2 MVI A,42

-S2CD3
2CD3 42 to 62
2CD4 C3 to .

(42H = side 1 + enable 2 MS clock. 62H = side 1 + RAM at Zero + Enable clock).

-L2CDA
2CDA MVI A,02

-S2CDB
2CDB 02 to 22
2CDC 32 to .

(02H = Side 0 + Enable Clock. 22H = Side 0 + RAM at Zero + Enable clock)

-L2CE1
2CE1 JMP 1D95
2CE4 STAX B

-S2CE4
2CE4 02 to 22
2CE5 09 to .

-GO

From A prompt SAVE 51 MOVCPMH8.COM

"Now run MOVCPM8.COM and SYSGEN to build a disk which will boot in a 400K Tandon Drive when using the HA-8-6 card.

"These patches modify the code written onto the Boot Track by turning on the "Ram at Zero" bit in the data bytes which are sent to the configuration port 362Q."

BIOS80 does not require any patches to work with H89's or with H8's using the HA-8-8 org-zero card. Preliminary indications are that this patch is incompatible with these other CPU's.

GOOD GUYS and BAD GUYS

The main purpose of this column is to let you know who you can feel confident dealing with. This includes criteria such as good service, fast service, good support, and good products. I will more readily write up a GOOD GUY than a BAD, because it is not my intention to drag someone through the mud. Only when I feel it is necessary, and I have received negative comments from many readers will an individual or company be written up as a BAD GUY.

This month I have many mentions for GOOD GUYS. Rather than devote a few paragraphs to each, I will give a brief mention of all of them, and why they have been nominated for GOOD GUYS, this being done to conserve space, and because in most cases there have only been one or two nominations for each.

If you have experienced exceptionally good service with an individual, company, or product, write in and let us know. On the other hand, if you have been burned, ripped off, or feel the product did not meet up to advertised specifications, write in. It is only through your feedback, whether positive or negative, that the rest of us will be able to make better choices. In this age of accelerated computer usage, development, sales, and products, it's more important than ever to know who to deal with, and what we can feel confident in purchasing.

GOOD GUYS--GREGORY MARSH was nominated by Ed Critz Jr for his mod to allow one to run up to 6 disk drives on the standard controller with a minor modification. Ed says it not only works great, but Gregory has provided good service (See NEW STUFF). He also has praise for Gregory's new STAT command that covers all the information one could want.

GOOD GUYS--Mark Rausher and Neil Rosner in a company called M & R ENTERPRISES/ 2242 E 28th St/ Brooklyn, NY 11229 have been nominated by Tom Jorgensen. M & R do graphics, typesetting and package design, and do it well. Tom, of SOFTWARE WIZARDRY Inc, said he has used their services to package his new software and is very pleased. They turn out jobs in short order, and the finished product is very professional in appearance and layout. The best thing, according to Tom, is their reasonable prices.

GOOD GUYS--Burton Hulland of Hulland Engineering was nominated by Welbrey Hill Jr. Burt's HDOS 0.1 "is so much less clumsy than Heath's HDOS even with Jim Teixeira's SYSMOD that it's not even funny. Secondly, the two bugs which I reported were responded to with the utmost dispatch. . ." Burt's product were mentioned in issues 15 and 16.

GOOD GUYS--NEW ORLEANS DATA SERVICE, the people who make the H8 music board marketed by Heath. Bruce Bennett told me they rewrote the software which supports the music board to make it CP/M compatible for him, and at no cost. I'm not sure what their policy will be as far as update pricing, but if you want to operate the music board for your H8 under CP/M, it may be worth your while to contact them. I have also had dealings with Dave of New Orleans Data Services, and was pleased with his support.

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APRIL issue deadline--March 15

Subscription Rates--\$20/Year (\$27 overseas Airmail)--current issues

Back issues available start with Volume 2, issue # 13. Volume 2 back issues available at \$1.25 each to issue #23, \$1.50 from #24 on.

PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday (Seventh Day Adventist).

NO COLLECT CALLS.

REALLY BAD GUYS--D. Knight (alleged name?) posing as Abileen HUG/ POB 1651/ Abileen, TX 79604. It appears that this guy, whoever, has used a Post office box, and "created" a temporary one man HUG to swindle vendors. He has ordered things very cleverly, asking for a 30 day purchase order payment date, and then nothing. In case guys scum like this don't know, that constitutes Mail Fraud, which is a Federal offense! I may have more to report on this in later issues. For now, if you get an order without cash from this area in Texas beware. As I printed long ago, Beware of post office boxes with no address. Should something like this happen, you have no way to find out who it is. This is even true when ordering from vendors which don't have a reputation established-- BEWARE!

HEATH GOOD GUY--Roger Svoboda of the Cincinatti store (513) 771-8850, was nominated by Bill Bechtold of Nebraska. ". . . is a 'company man'; but he is first of all, interested in solving a customer's problems, and he does not want to know your bank balance before he will talk to you. It is well worth the price of a phone call to talk to him if I am not sure which way to go on a purchase, or want immediate delivery of something. If my local store can only "order it for you", or "check into it", etc., I simply call Ohio, and usually have whatever I need within 2-3 days."

CLASSIFIEDS

FOR SALE--16 K RAM boards (5) in mint condition, 3 brand new, \$175 each. Two 8K RAM boards in excellent condition for \$75 each. Z89 16K upgrade kit, brand new in factory box (2) for \$125 each. Two Siemens disk drives FDD-100-5B004 (flippy) DD or SD in mint condition at \$250 each. Contact Hank Guell/ 246 Skyline Lk. Dr./ Ringwood, NJ 07456/ (201) 839-9071 after 9PM EST.

FOR SALE--DG Electronics 32K memory board for the H8 computer, in perfect condition for \$179. Richard Calhoun/ 1738 11th st/ Langley AFB, VA/ 23665/ (804) 827-5059 after 5PM.

FOR SALE--Four Heath 8K RAM boards, \$55 each. Contact Jim Scott/ 1724 Blakemore Rd/ Richmond, VA 23225. Phone (804) 232-2925.

FOR SALE--Factory sealed Heath SuperCalc software packages HSC-817-1 for \$145 each, prepaid to "International Software", or \$150 via UPS COD. This includes shipping within USA. John Dunning of International Software/ 1395 Main St/ Waltham, MA 02154/ (617) 899-7244.

FOR SALE--10 Mb CDC Hard Disk (HAWK) for \$1000. Heads fly, but no way to test it on H8 or H89. Contact R Zuck/ 624 S. Columbus St/ Alexandria, VA 22314. Phone (213) 536-3066 (calls will be forwarded).

FOR SALE--H8 8080 CPU Board \$100, Heath 16K RAM board, factory assembled \$170, Heath 8K RAM boards- 3 for \$170 or \$60 each. All 100% functional and for the H8 computer. Contact Jerry Nyhart/ 3017 Corthouse Dr/ W. Lafayette, IN 47906. Phone (317) 463-3883 or 255-5950.

WANTED--Heath H9 video terminal. Contact Haze Huntmacher/ W 11428 Tumbleweed/ Nine Mile Falls, WA 99026